

Advanced Characterization and Analysis of Random Telegraph Noise in CMOS Devices



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1 Introduction

Random telegraph noise (RTN) is a well-known phenomenon in electron devices, which has been extensively studied for more than 30 years [1–3]. Since its first observation in 1984 by Ralls et al. [4], hundreds of publications have dealt with RTN, with an explosion in the last years due to the relevance that it has gained in the most recent nodes, in both bulk and SOI technologies [5–8].

RTN is a consequence of trapping/detrapping of charges in/from defects (or, equivalently, traps) in the gate dielectric and/or its interfaces, which lead to sudden and random changes in the device current (see an illustrative example in Fig. 1). When considering the drain current in a MOSFET, these changes are associated with shifts in the threshold voltage (V_{th}) of the device induced by the trapping/detrapping of charges. These random and transient changes in the device V_{th} can be actually a source of device time-dependent variability (TDV). Actually, for extremely scaled technologies, the RTN contribution to the variability of CMOS logic devices and circuits may overtake that of random dopant fluctuations (RDFs) as its dominant source [9].

The transient variability introduced by RTN in the threshold voltage of the devices will be translated into transient shifts of the circuit electrical parameters, negatively impacting the circuit performance and reliability. In logic circuits, RTN can seriously influence the energy-delay trade-off in digital applications [10, 11]. In memories, RTN causes variable retention times in DRAM [12, 13], bit cell failure in SRAM [14], and in NAND flash memories, since its effect becomes more pronounced as the number of bits per cell increases [15]. SRAM cells

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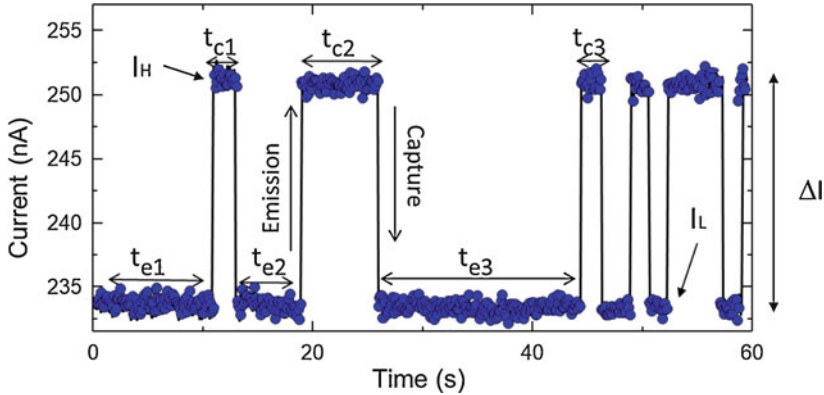


Fig. 1 Typical two-level RTN trace (obtained through simulation) caused by the charge trapping/detrapping in/from a single defect located in the transistor gate oxide. Background Gaussian noise has been added to the RTN signal, to mimic the real situation

fabricated in SOI technologies are also sensitive to RTN [16, 17], which can cause intermittent errors contributing to bitcell failure. Note that the ultrathin body and buried oxide fully depleted silicon-on-insulator (UTBB FD-SOI) technology offers new advantageous design opportunities thanks to its very efficient body-biasing capabilities [18], introducing the body of the device as a second gate. However, from an RTN perspective, both front and back gate (FG and BG) of the device could act as RTN sources. Still in the memory field, emerging memory architectures, such as ReRAM [19, 20], can also be affected by RTN. In these memories, a gate voltage applied to a metal–insulator–metal (MIM) or metal–insulator–semiconductor (MIS) structure may reversibly change the resistance of the dielectric between a high-resistance state (HRS) and a low-resistance state (LRS). RTN can be observed in the current through the device, especially in its high-resistance state (HRS) [21], where it can cause very wide current fluctuations, potentially inducing read failures [22]. Though the defects and physics behind RTN in this kind of memories probably differ from those based in MOSFETs, RTN coming from all types of devices are qualitatively equivalent.

From the above, it can be concluded that, with current CMOS technology downscaling, the time-dependent variability introduced by RTN must be taken into account in the design of memories, digital and analog VLSI-integrated circuits (ICs), to implement reliability-aware circuits and systems. To fill the gap between the device and circuit levels, the TDV introduced by RTN must be introduced into the device compact models. Then, the development of physics-based compact models that accurately describe the observed shifts of the electrical parameters of the devices is a must. Moreover, suitable and accurate model parameters extraction procedures must be developed in order to account for the specific characteristics of the technology under study. In this regard, because of the particular features of RTN, which will be described in the next section, the characterization and data analysis needed for RTN parameter extraction in advanced devices are challenging.

1.1 RTN Phenomenology

As described previously, RTN is related to the trapping/detrapping of charges in/from traps in the gate oxide and/or its interfaces. A detailed explanation about the physics related to the RTN phenomenon can be found in chapter “Oxide Trap-Induced RTS in MOSFETs” of this book [23]. In MOSFETs, these trapping events lead to changes in the threshold voltage of the device [24]. This is shown in Fig. 2, where TCAD simulations of MOSFETs with trapped charges at a location of the channel interface are shown. Clearly, the charges introduce changes in the potential profile of the device (Fig. 2a), which are translated into a shift of its I_D – V_G transfer characteristics (Fig. 2b). Less current is driven by the device for a fixed V_G , which can be interpreted as a shift in V_{th} . If the charges are detrapped, the initial characteristics would be recovered, i.e., the initial V_{th} . As a consequence, when a constant voltage is applied, RTN is manifested as discrete jumps in the drain current between two levels, as defined by the trap state (neutral or charged). As an example, Fig. 1 schematically shows the RTN signal associated to the simplest case, which corresponds to the presence of only one defect in the device. Experimentally, RTN signals will be always accompanied by the background experimental noise, so that the current/threshold voltage levels could not always be precisely defined (see, e.g., Fig. 4a). The drain current switches between two current levels, I_H (when charges are detrapped) and I_L (when charges have been trapped), separated by a distance ΔI . The times t_e and t_c are the times that a charge needs to be emitted or trapped, respectively, which have been observed to be statistically distributed, following an exponential distribution with averages τ_e and τ_c [25]. Three parameters are required to describe a two-level RTN signal like the one shown in Fig. 1; the mean emission (τ_e) and capture (τ_c) times and the magnitude of the current shift (ΔI). The last parameter, ΔI , can be obtained directly from the RTN trace. These parameters

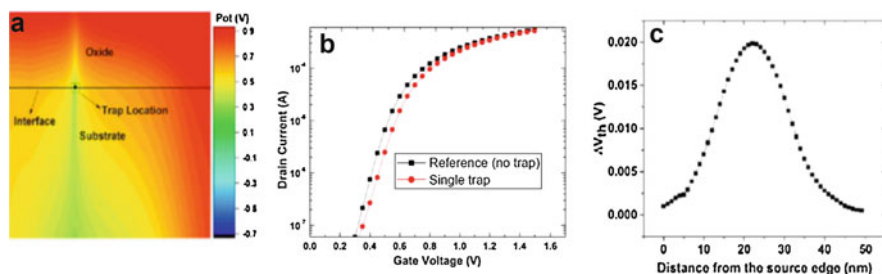


Fig. 2 2D TCAD simulations of a MOSFET, when a single charged trap is considered at its interface: (a) Potential profile. (b) I_D – V_G of the device when the trap is neutral (black squares) or charged (red circles). Charge trapping leads to a change of the potential in the channel and a shift of the transfer curve towards the right. This shift means smaller currents for the same bias, which can be interpreted as an increase of the threshold voltage. (c) The threshold voltage shift depends on the location of the trap along the device channel. In this 50-nm long channel device, the impact is maximum when it is located close to the center of the channel

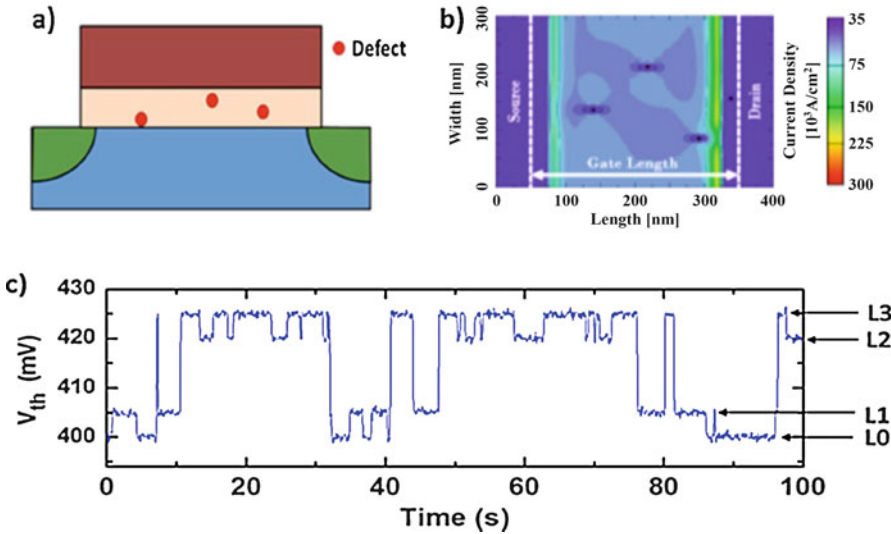


Fig. 3 (a) Schematic front view of a MOSFET with three defects located at different depths in the gate oxide and location along the channel. (b) TCAD simulation of the current density in a device with three defects. Charged defects locally affect the conduction across the channel. (c) Example of a typical RTN trace (obtained through simulation) in a device with two defects. The charging and discharging of the defects lead to multiple levels in the RTN (L0–3). Background noise could mask these shifts

are strongly related to the physical characteristics of the defect (i.e., energy level, location in the oxide and along the channel...). As an example, Fig. 2c shows TCAD simulation of the threshold voltage shift when charged traps are located at different locations along the channel. In this short channel device, charges close to the center of device have a larger impact, whereas those close to the source/drain have a smaller effect [26]. Consequently, depending on the position of the trap along the channel, a different ΔI will be introduced in the device drain current. Other authors have also shown the trap impact dependence with the trap position and the random dopant distribution (RDD) [27] and it is also discussed in chapter “RTN and Its Intrinsic Interaction with Statistical Variability Sources in Advanced Nanoscale Devices: A Simulation Study” of this book [28].

Two-level RTN traces are related to a single defect but, actually, in a device more than one defect can be active, located at different positions along the interface and/or oxide depth (see Fig. 3a). Figure 3b shows the current density in a device where three defects are present and charged. It is evident that each of these defects will lead to a different ΔV_{th} (or, equivalently ΔI), since they are located at different positions at the interface, and could also be characterized by different time constants (τ_e , τ_c , and ΔI) [29]. Then, trapping/detrapping events in the defects will lead to multilevel RTN traces. In the ideal situation (absence of noise, of interaction among traps, ...), the number of levels is determined by the number of defects in the

device: for N defects, 2^N levels could be distinguished. As an example, Fig. 3c shows an RTN trace with four levels, which results from the presence of two defects in the device. However, the number of defects in a device inversely depends on the device area, so that in advanced nodes this number will be typically small [30].

From the discussion above, we can conclude that RTN is an intrinsically stochastic process that introduces random and transient V_{th} changes in the device. Moreover, it will change from device to device because:

- The number of defects per device in a set of devices is statistically distributed. It is commonly assumed that this number follows a Poisson distribution [29].
- The drain current change (ΔI) or, equivalently, the V_{th} shift (ΔV_{th}) associated with each defect is exponentially distributed [31].
- τ_e and τ_c are widely distributed, expanding over several time decades, ranging from μs to seconds. Note, however, that this estimation has been done taking into account typical experimental windows, so that smaller and larger time constants cannot be ruled out.

Taking into account the characteristics of the phenomena, it becomes clear that (1) a statistical characterization of the phenomenon becomes mandatory (many devices, long-lasting traces) and (2) advanced automatic analysis methodologies are needed, in order to process the huge amount of data that will be generated from such characterization. These two issues will be the main focus in this work.

Before going into RTN characterization and analysis, it must be noted that a similar physical framework has been proposed to describe the phenomenon of bias temperature instability (BTI), one of the most relevant aging mechanisms in MOSFETs that limits the circuit reliability in current technologies. As RTN, BTI is observed in the device electrical characteristics as a change in the device threshold voltage. However, in the case of BTI, two separate components can be distinguished in the threshold voltage shift, i.e., a permanent component (related to the charging of defects that are not discharged in the experimental window) and a recoverable component (related to the discharging of a defect when the gate bias is decreased or removed). The observation of RTN and/or BTI is strongly dependent on the operation conditions (voltage and temperature) of the device [32]. RTN and BTI effects can be observed simultaneously, the first one as fast increase/decrease events in the device current, the second one as slower or permanent changes in the current. Actually, nowadays it is believed that RTN is an important factor in the understanding of BTI reliability issues in advanced CMOS technologies [33]. In this regard, it has been proposed that RTN and the recoverable component of BTI are very likely related to the same kind of defects [34]. Then, (1) all defects that contribute to the BTI recovery can also become spontaneously charged, leading to an RTN signal, (2) most RTN defects would also contribute to BTI recovery, (3) the distributions of RTN and BTI characteristic parameters are similar for the two phenomena, and (4) both RTN and BTI defects are volatile, meaning that they can disappear and reappear [34]. A very important consequence of the similarities between RTN and BTI is that they must be analyzed and guardbanded

against together. In this direction, models that are able to describe both phenomena simultaneously have been proposed [35]. Though RTN and BTI are closely related, in this work, RTN will be the main focus.

2 The Weighted Time Lag Method

In this section, an advanced methodology suitable for the extraction of the RTN current/threshold voltage levels from RTN traces will be described in detail. The method has a wide applicability, provided that the signals to be analyzed show the typical RTN signature. In addition, it overcomes the detection limitation introduced by background noise, which can impede the precise detection of levels (i.e., the defect parameters), since their contribution can be hidden. The method here presented for the RTN analysis extends the time lag plot (TLP) procedure [36], being easily implementable and robust even when the background noise is large. The method will be first explained and illustrated using as example a numerically generated RTN caused by only one defect. After that, the presented method is applied to identify the contribution of defects associated to experimental multilevel RTN.

To illustrate the method, numerically generated RTN signals have been considered. Then, Monte Carlo simulations of two-level RTN waveforms were performed by considering two-state Markov processes [37]. The black line in Fig. 4a shows an example of a simulated current RTN without noise with levels $I_L = 3$ nA and $I_H = 5$ nA. Assuming that the responsible defect of this RTN is occupied at I_L and empty at I_H , its mean capture and emission times were chosen to be $\langle\tau_c\rangle = 10$ s and $\langle\tau_e\rangle = 6.6$ s, respectively. Dots in Fig. 4a show the waveform obtained when a Gaussian noise with standard deviation $\sigma = 0.9$ nA (to represent the experimental background noise) is added to the RTN. Due to the large unfavorable ratio between the noise and the RTN amplitude, an accurate direct I_L and I_H identification is unfeasible from Fig. 4a. The RTN histogram (dots in Fig. 4b) cannot help, because the two peaks associated with I_L and I_H are hidden by the background noise [38]. Another solution is to draw the TLP (Fig. 4c), which is constructed by plotting the i th point of the RTN on the x -axis and the $(i + 1)$ th point in the y -axis for the full RTN trace [36]. If the background noise is low compared to the RTN amplitude, using the TLP, the RTN levels can be identified as populated regions in the diagonal, while populated regions outside the diagonal are related to the transitions between states [36] (these regions are indicated with crosses in Fig. 4c). However, in the TLP constructed from the RTN trace in Fig. 4a, these regions again overlap because of the background noise.

The weighted time lag (w-TLP) method [39] tries to extend the TLP by minimizing the effect of the noise in the RTN, so that a more accurate defect parameters extraction is possible. The starting point of the w-TLP method is a point of the TLP with coordinates (I_i, I_{i+1}) . For this point, we define the φ_i function as

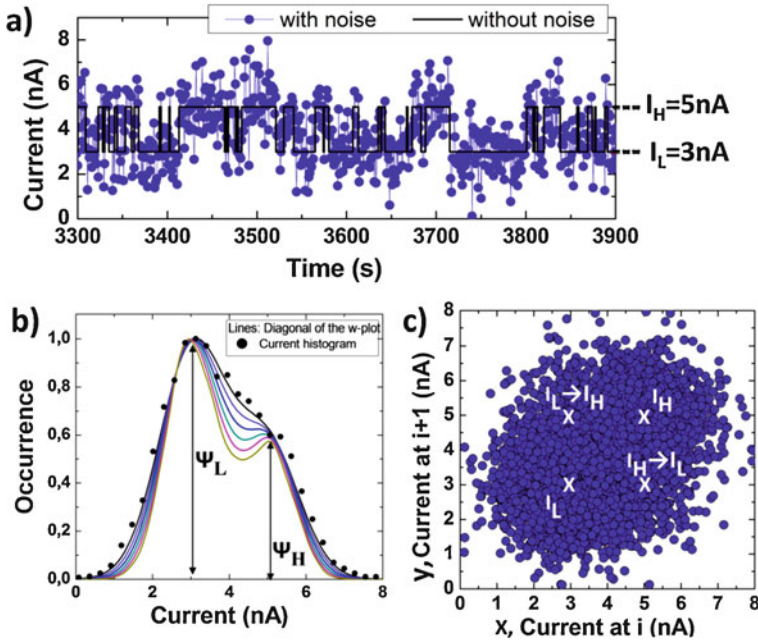


Fig. 4 (a) Monte Carlo-generated RTN signal with and without noise; the sample rate considered was 1 s and the number of points in the RTN is 10,000. (b) Dots indicate the current probability distribution obtained directly from the RTN signal in (a); lines correspond to the diagonal of the w-TLP (Fig. 5c) obtained for different values of α , which is the standard deviation of φ_i (Eq. 1). (c) TLP of the RTN; the background noise hides the current levels and transition regions (marked with crosses)

$$\varphi_i(x, y) = \frac{1}{2\pi\alpha^2} \exp\left(-\frac{[(I_i - x)^2 + (I_{i+1} - y)^2]}{2\alpha^2}\right) \quad (1)$$

where x and y are the coordinates of the space where the TLP is considered. Note that φ_i is a normal bivariate distribution with standard deviation α and correlation coefficient 0. Then, $\varphi_i(x, y)$ represents the probability that the point with coordinates (I_i, I_{i+1}) corresponds to a level or to a transition in the a location (x, y) of the TLP space. Figure 5a shows the 2D plot of $\varphi_i(x, y)$ in log scale for a single point of the TLP. After the φ_i definition, we define the weighted time lag function Ψ as:

$$\Psi(x, y) = K \sum_{i=1}^{N-1} \varphi_i \quad (2)$$

being K a normalization constant chosen so that the maximum value of Ψ is 1 and N the number of points in the RTN. If Ψ is plotted for few points (Fig. 5b), two local maximums are roughly defined whose values are closer to I_L and I_H .

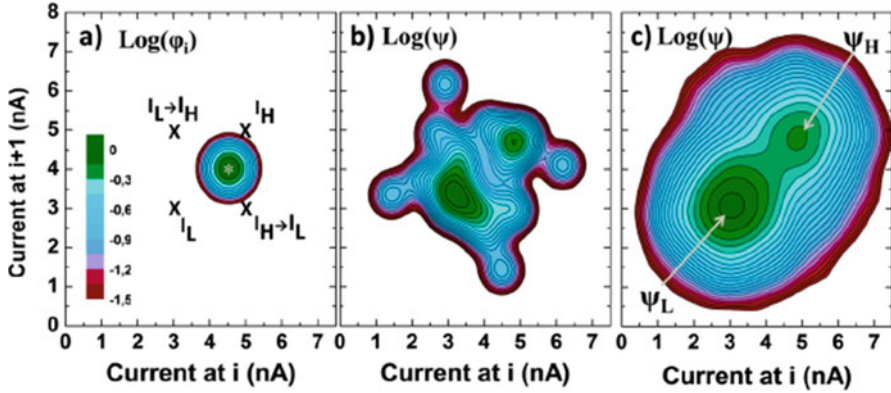


Fig. 5 (a) Representation of $\text{Log}(\varphi_i)$, normalized to its maximum value, obtained for one point of the noisy RTN in Fig. 4a. The position of this point in the TLP is marked with * and the levels and transitions regions with crosses. (b) Representation of $\text{Log}(\Psi)$ using the first 30 points of the RTN and (c) the full RTN trace

To understand why these peaks are revealed, we have to note that the contribution to Ψ of each point of the TLP is weighted by the distance between the position of this point and (x, y) . Therefore, the function Ψ takes higher values in the most populated regions of the TLP. This is more evident in Fig. 5c where the function Ψ is plotted for all the points of the TLP. In the following, we will refer to the plots of the type of Fig. 5c as weighted TLP (w-TLP). In the diagonal of the w-TLP of Fig. 5c two well-defined local maximums can be detected (Ψ_L and Ψ_H) in the positions 3.05 and 4.89 nA. These values correspond to the two levels of the RTN, which are very close to the nominal $I_L = 3$ nA and $I_H = 5$ nA values considered to generate this signal (Fig. 4a). Then, the construction of the function Ψ is valid to detect levels of the RTN that cannot be determined from conventional methods, such as the RTN histogram (Fig. 4b) or the TLP (Fig. 4c), when the background noise is relevant. The ratio τ_e/τ_c , which in most of the cases is enough to obtain the relevant physical information of the defect [40], can be easily calculated from the w-TLP by the evaluation of the ratio of the two local maximums found in the w-TLP (Ψ_H/Ψ_L). In this case, we obtain $\Psi_H/\Psi_L = 0.63$, again close to the nominal value imposed during the generation of the RTN signal ($\tau_e/\tau_c = 0.66$).

In the weighted time lag method, the chosen value of α (the standard deviation of φ_i), which is the only fitting parameter, is a key point to correctly identify the levels of the RTN. Lines in Fig. 4b show the cross section through the diagonal of the w-TLP obtained when Ψ is constructed using different α values. For high α values, only one peak (as in the case of the histogram, dots in Fig. 4b) appears, and the two peaks are only revealed when α is decreased. However, if α is very low, this procedure fails because each point only contributes to Ψ in a region very close to it, leading in the limit, to a conventional histogram with a small bin size. Then, the proper selection of α is crucial to correctly determine and identify the defect in the RTN. This point will be further discussed later.

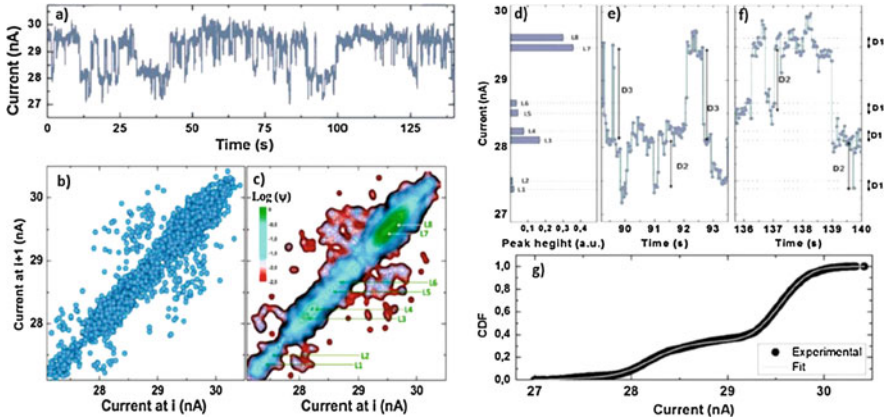


Fig. 6 (a) Experimental multilevel RTN in the drain current of a pMOS transistor. Conventional TLP (b) and w-TLP (c) obtained from the RTN of (a). (d) Position and height of the peaks detected in the diagonal of the w-TLP in (c). (e, f) Zooms of the experimental RTN trace in (a). Switchings of the three defects are indicated. D2 and D3 are evident but those of D1 cannot be clearly appreciated because of the background noise. (g) Experimental cumulative distribution function of the RTN in (a) and the fitting obtained from the data in (d)

The w-TLP method can be applied to multilevel signals without any change. To show this point, we considered an experimental RTN signal with a sample rate of 34.2 ms in the drain current of a pMOS transistor when it is biased with a gate voltage of -0.6 V and drain voltage of -200 mV during 140 s. The obtained current trace (Fig. 6a) shows a clear RTN where at least two discrete steps can be detected, which indicates that at least one defect is active in this device. However, a large background noise is present, making the accurate identification of the resulting current levels difficult. Moreover, other defects that could provoke smaller current steps could be hidden by the noise.

Figure 6b shows the TLP of the RTN plotted in Fig. 6a. Note that, when using the TLP, the current levels in the signal are difficult to distinguish. However, with the w-TLP (Fig. 6c), eight local maximums (L1–L8) can be detected. This suggests the existence of eight current levels in the RTN, and therefore at least three active defects in the device. The position and height of the eight current levels observed in the w-TLP are plotted in Fig. 6d. Considering the position of the current levels, three defects with different switching amplitudes have been identified (D1, D2, and D3). The current shifts between the levels associated with D2 and D3 are large enough to detect the transition regions in the TLP and w-TLP (regions outside the diagonal in Fig. 6b, c), and also can be detected in a zoom of the RTN (Fig. 6e, f). However, as in the case of the generated signal in Fig. 4, the small current shift associated with D1 compared to the amplitude of the background noise impedes the correct identification of D1 directly from the RTN.

Finally, the continuous line in Fig. 6g shows the cumulative distribution function (CDF) of the experimental RTN (symbols), which can be well reproduced using the

data of Fig. 6d. This fitting is done by associating a normal distribution with each of the current levels detected in the w-TLP. The mean value of the normal distributions corresponds to the current level position; their height corresponds to the relative peak height obtained from the w-TLP. The standard deviation ($\sigma = 0.22$ nA) is chosen to be the same for all the current level distributions since it is related to the experimental background noise. The good fitting of the experimental data allows concluding that the weighted time lag method is an efficient procedure to identify defects in the device and obtain an accurate description of the resulting RTN. Another example of RTN signals analysis using the w-TLP method can be found in chapter “Systematic Characterization of Random Telegraph Noise and Its Dependence with Magnetic Fields in MOSFET Devices” of this book [41].

3 RTN Experimental Characterization Challenges

Precise RTN characterization in modern CMOS technologies is a key point towards the design and fabrication of reliable CMOS integrated circuits (ICs). However, the particular features of RTN traces pose important requirements to the experimental setup needed for such characterization. As can be concluded from the previous section, low-noise systems are required in order to discern the presence of defects whose contribution to the device current could be hidden by the unavoidable background noise. But other considerations are also to be accounted for. On the one hand, transients that are characterized by time constants in the microsecond range may appear. Then, systems with a large time resolution must be used to capture those fast defects. On the other hand, due to the stochastic nature of these phenomena, a statistical characterization is mandatory, so that a large number of devices must be measured to obtain statistically relevant results. In the next subsections, we address some of the approaches that can help to fulfill these requirements.

3.1 Ultrafast Measurement Techniques

Usually, standard characterization equipment has been commonly used to characterize RTN, such as semiconductor parameter analyzers (SPAs), with typical measurement time resolution in the ms range. To improve the time resolution, high-cost equipment can be used, such as the ultrafast modules that can be added to a standard SPA (as, e.g., Keithley 4225-PMU ultrafast I–V module [42] or Keysight B1530A waveform generator/fast measurement unit [43]). Here, we propose a low-cost, custom-made experimental setup to measure RTN, which provides a higher time resolution than that available in standard characterization equipment and comparable to the one provided by more expensive systems (in the microsecond range). As will be shown, when the results with this setup are analyzed with the w-TLP method, additional and relevant information about the RTN phenomenon can be obtained. As an application example, the setup has been used to characterize an

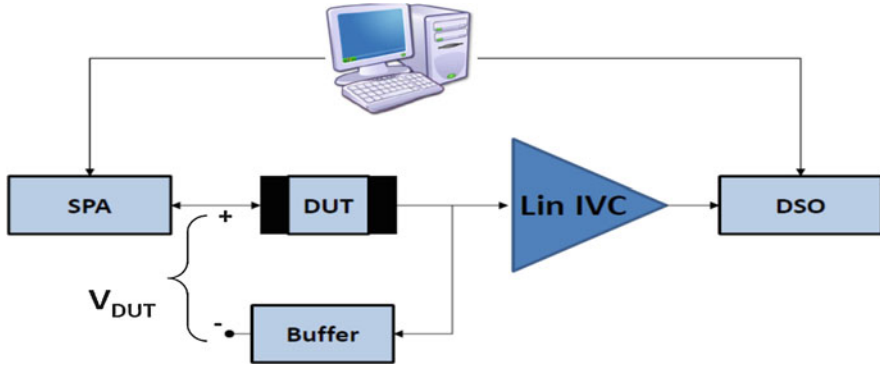


Fig. 7 Schematics of the experimental setup developed to measure RTN signals with large time resolution. V_{DUT} is the voltage drop across the DUT

RTN signal coming from RRAM devices. Though RTN signals can be observed in both resistance states (HRS [44] and LRS [45]), this work is focused on the analysis at HRS, because RTN becomes more relevant in the case of larger dielectric resistance values [46], i.e., lower currents through the dielectric.

$5 \times 5 \mu\text{m}^2$ Ni/HfO₂ (20 nm)/Si RRAM devices were characterized. In order to get a stable conduction, the devices were first subjected to 30 resistive switching (RS) cycles (where the first cycle corresponds to the forming process [47]), changing successively the dielectric conductivity between a high- and a low-resistance state, fixing a current limit of 10 μA when switching to the LRS [47]). After this initial cycling, when the device was at HRS, several constant voltages were applied until a clear RTN signal was found. To allow comparisons, the RTN signal was measured simultaneously with SPA and the new setup. Figure 7 shows a schematic of the experimental setup developed to measure RTN with large temporal resolution. The voltage at which RTN appears was applied to one of the terminals of the DUT by the SPA (Agilent 4156C in our case), which also allows measuring the current through this terminal. The other terminal of the DUT was connected to a current-to-voltage converter (Lin-IVC), which is connected to a digital storage oscilloscope (DSO) in order to register data at different time scales. This second DUT terminal is also connected to a high impedance buffer, so that a precise value of the voltage drop across the DUT can be measured. By changing the timescale of the DSO, the current through the DUT can be easily measured with different time resolutions, so that fast events can be captured.

Figure 8 shows the flowchart of the RTN measurement process. Once a suitable value of voltage has been selected, the HRS current is measured as a function of time. While the SPA is continuously measured during a predetermined time, several oscilloscope captures are registered using different time scales. When the SPA measurement finishes, the sequence defined in Fig. 8 starts again. In this work, about 325 SPA sequential measurements were done and 21 oscilloscope captures were obtained for each SPA measurement.

Fig. 8 Flowchart of the RTN measurement process once a RTN signal is detected at a particular voltage. RTN signals are simultaneously measured with the SPA and the developed setup. The time scale of the DSO is changed during the measurement

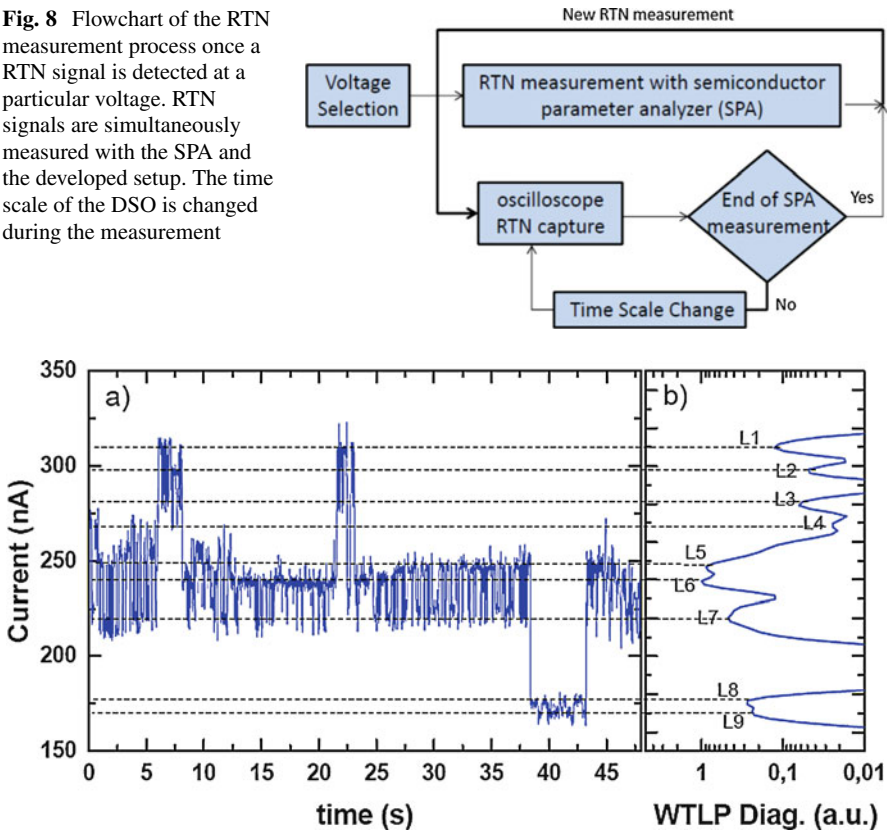


Fig. 9 (a) Typical multilevel RTN signal measured with a semiconductor parameter analyzer. Applied voltage = 1.25 V, step time ~6 ms, and number of measured points 8000. (b) Current levels obtained by using the w-TLP method

In Fig. 9a, a typical multilevel RTN signal measured with the SPA in a RRAM, with a time resolution of 6 ms, is shown. At naked eye, five different current levels are evidenced. However, by applying the w-TLP method to analyze this RTN signal, more levels are encountered, concretely nine different levels (L1–L9) (Fig. 9b). Figure 10 shows three oscilloscope captures of the same signal using several time scales, or equivalently time resolutions of (a) 40 μ s, (b) 20 μ s, and (c) 1 μ s, recorded simultaneously at different time slots. In the three cases, some charge trapping and detrapping events are detected with switching times lower than 6 ms. These fast current fluctuations are not detected with the SPA (Fig. 9). Figure 10 shows that some current levels coincide with those in Fig. 9b (L2, L3, L5, L6, and L7). However, in Fig. 10b and Fig. 10c, there is a new level (L10) that does not appear in Fig. 9a, because of the fast transitions between states.

To show these points clearly, the w-TLP method was used. In Fig. 11, the w-TLPs corresponding to the RTN signals measured with the SPA and the oscilloscope

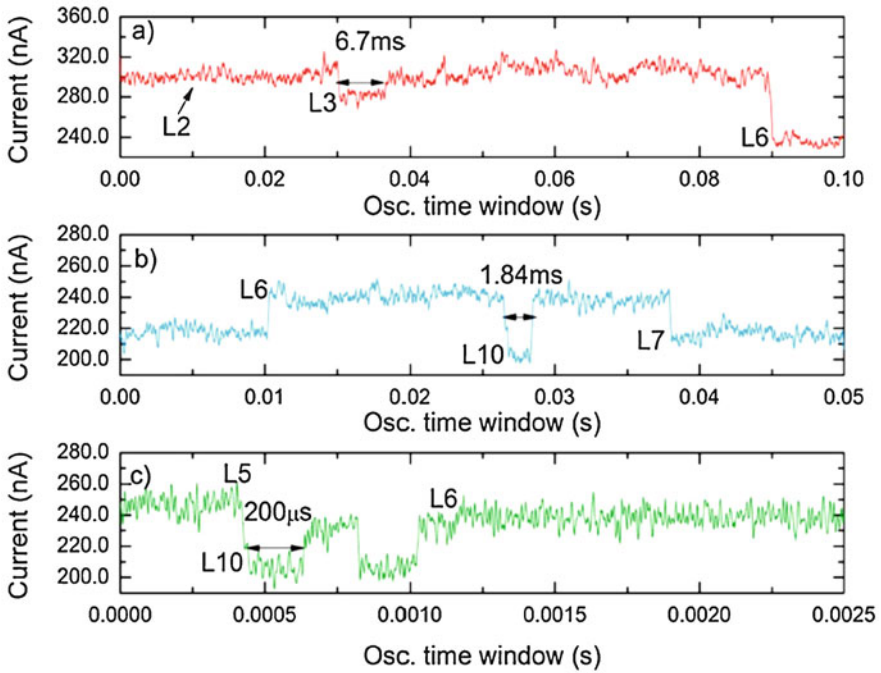


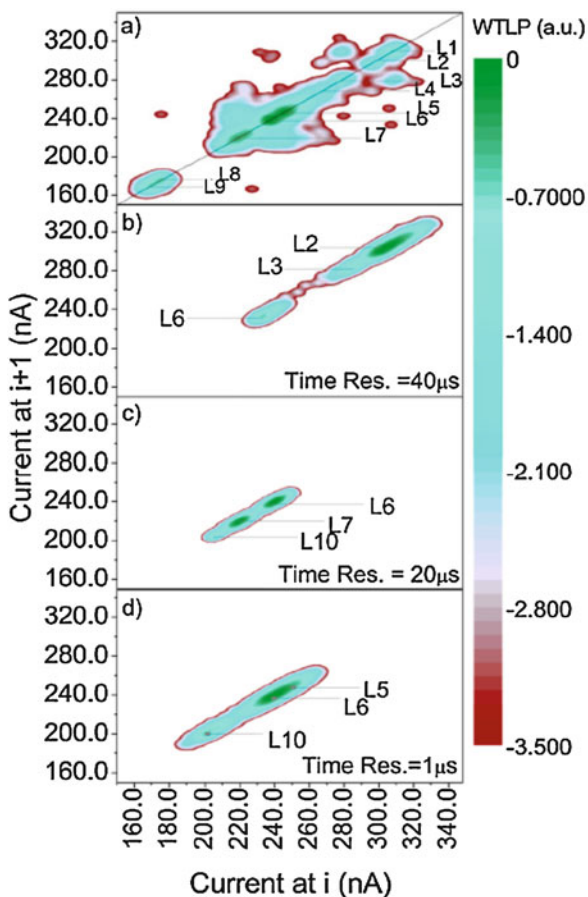
Fig. 10 Traces captured with the oscilloscope in the interval of: (a) 7.26–9.68 s of the SPA measurement shown in Fig. 9a with 40-μs time resolution. (b) 9.68–12.15 s, with a time resolution of 20 μs. (c) 19.5–21.93 s with a time resolution of 1 μs

are obtained. Figure 11a shows the w-TLP of the complete RTN signal measured with the SPA presented in Fig. 9a. A profile of the diagonal is plotted in Fig. 9b where all the detected levels are sequentially numbered. From the analysis of the w-TLP diagonal, nine peaks and therefore nine current levels (labeled as L1–L9) are obtained. Note that the number of current levels as determined by the w-TLP method is larger than that encountered at first sight. The guidelines plotted in Fig. 9 help to observe a good match between the peaks in the diagonal of the w-TLP and the current levels in Fig. 9a.

Figure 11b–d shows the w-TLPs of the three RTN signals captured with the DSO, and, as in the case of Fig. 11a, the existing traps levels are identified. Comparing these figures with Fig. 11a, we can recognize levels 2, 3, 5, 6, and 7. But from Fig. 11c, d, a new trap level at ~200 nA, labeled L10, appears, as mentioned before. This latter state does not appear in the SPA measurement because of the equipment resolution. However, the larger time resolution of the developed setup highlights its presence and it can be clearly visualized in the corresponding w-TLP.

In summary, combining high time resolution equipment and accurate parameters extraction, a precise characterization of the RTN current fluctuations observed

Fig. 11 w-TLP method applied to (a) the RTN measured by the SPA (Fig. 9a). The line shows the diagonal of the plot, whose maxima correspond to the RTN current levels. Nine trap levels are detected. The RTN captured with the oscilloscope with time resolutions of (b) 40 μ s (Fig. 10a), three trap levels appear; (c) 20 μ s, with three trap levels appear (Fig. 10b); (d) the larger time resolution (1 μ s), showing three trap levels (Fig. 10c)



in devices can be carried out using the w-TLP method, the RTN levels can be accurately identified.

3.2 Statistical RTN Characterization: Array-Based Solutions

The stochastic nature of the RTN phenomenon requires the investigation of a very large number of samples to evaluate the statistical parameters required for a reliable estimation of its impact on the circuit functionality. The customary used approach of direct testing of devices on a wafer (i.e., direct connection of the measurement instrumentation to the contact pads of the device in the wafer on a probe station) would then become unfeasible. On the one hand, direct measurement of RTN traces with typical durations of 100–1000 s in a large number of samples will require impractical measurement times. The automation of the full characterization process

would partially overcome this problem, but still the testing time would be very large. One solution is the parallelization of the measurement procedure (i.e., several devices at a time), but this requires a carefully designed wafer layout (e.g., devices could not share contact pads) and/or multiple measurement equipment, increasing the cost of the setup. Then, the reduction of the RTN testing time is a challenge. On the other hand, individual, isolated devices in a wafer would be very expensive in terms of silicon area, since most of the area of the wafer would be occupied by the contact pads of the devices. To reduce the silicon area, a feasible solution is the fabrication of ICs that contain arrays of devices, where device contact pads are shared and a specific circuitry for device selection is included. Such kind of device arrangement has been proven successful for the evaluation of process variability and aging of devices [48, 49]. This access capability brings the advantage that the total silicon area used will be considerably reduced, as compared to the probe station approach. Moreover, with this arrangement, automatic characterization of thousands of DUTs can be carried out, reducing the required testing time. It must be mentioned, however, that though the simultaneous biasing of several devices is possible (to reduce the stress time when performing reliability tests), usually only currents through one device can be measured at a time.

However, though conceptually simple, designing a transistor array for this type of measurement is not a simple task. A set of requirements must be fulfilled to carry out a proper characterization of the phenomena. For example, when current is flowing through a DUT, a voltage difference appears between the DUT terminals and the externally applied voltage on the corresponding IC pad. This voltage drop is caused by the series resistance of the chip metal paths, the access circuitry, and the chip pads. Therefore, for accurate measurements, force-and-sense techniques are necessary, meaning that independent force-and-sense paths are required to access the DUT terminals where current is flowing. With this access structure, the voltage at the DUT terminal is sensed through the sense path, while the external instrumentation adjusts the force voltage, until the desired voltage value is set at the DUT terminal. Also, calibration techniques are required to compensate the leakage currents coming from the access circuitry that are added to the current of the measured transistor.

Recently, several examples of this device arrangement have been presented, with different advantages and/or disadvantages, mainly developed for the statistical characterization of process variability and device aging, but only few of them have been used for RTN characterization [50, 51]. For example, the chip presented in [50] contains 96×18 cells, each including 48 DUTs and an A/D converter that serially digitizes the current of each DUT. The chip, which contains devices with various channel doping and lengths (with $W = 80$ nm and $L = 30$ or 86 nm), was fabricated in a 28-nm HKMG bulk process. Due to the selected architecture, no force-and-sense scheme is necessary, although a calibration is performed for leakage compensation.

The recently developed ENDURANCE transistor array chip [52], fabricated on a 65-nm technology, contains 3136 MOS transistors (nMOS and pMOS) of different geometries, to allow statistically significant results for electrical characterization,

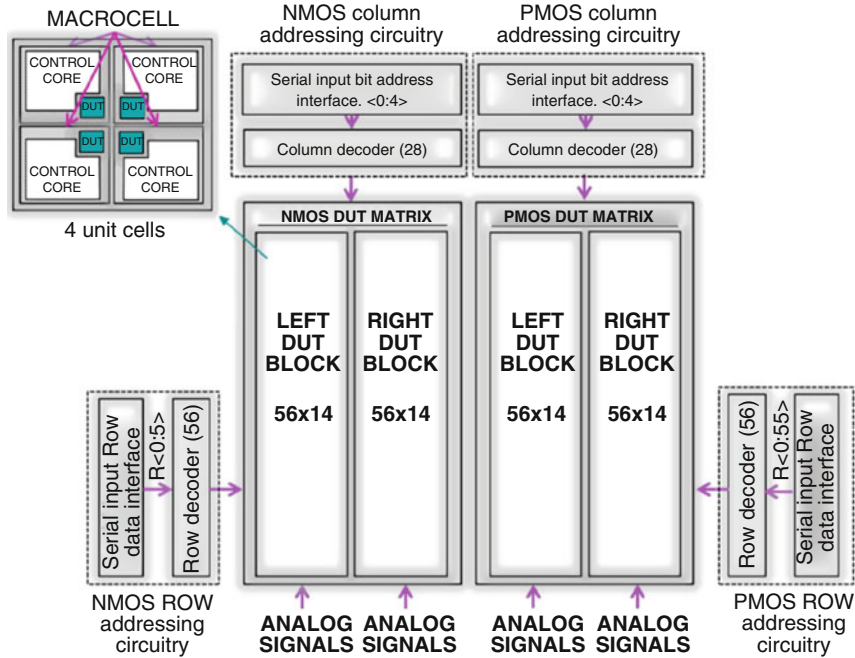


Fig. 12 Architecture of the ENDURANCE chip

not only of RTN, but also to evaluate process variability, BTI and HCI aging using a dedicated IC (Fig. 12). The chip has demonstrated a great versatility, providing parallelized accelerated aging tests with accurate timing, and a force and sense technique that compensates IR drops. Using this array chip, a full RTN characterization can be automatically done. As an example, Fig. 13 shows a set of several current traces displaying RTN effects. This figure demonstrates that the system is capable of capturing current fluctuations associated with charge trapping/detrapping in/from defects in the analyzed devices, with different emission and capture times. Multilevel signals, corresponding to more than one defect in the transistor, can also be observed. The control logic included in the chip allows the automation of the tests, partially reducing the experimental time needed for the characterization of a large number of devices [53].

4 Other Applications of the w-TLP Method

In Sects. 2 and 3, we have shown the potentiality of the w-TLP method for the evaluation of the current levels in the RTN signals, which could come from either RRAM or MOSFETs. In this section, we will describe two other application

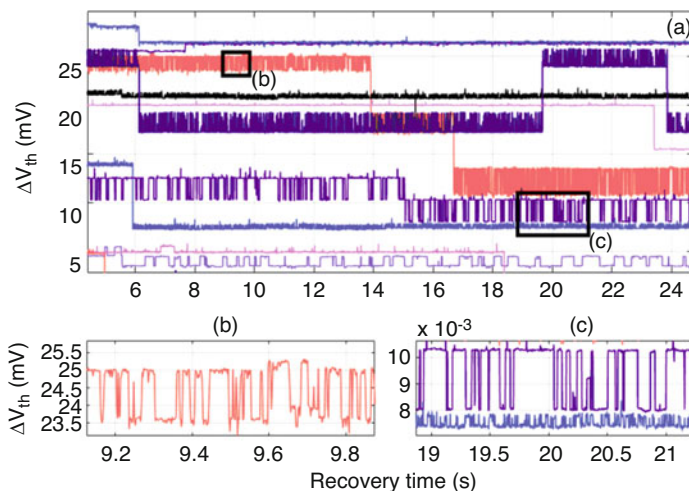


Fig. 13 (a) Examples of RTN tests on seven DUTs using the ENDURANCE chip [52]. (b, c) Zooms of RTN traces

examples of the w-TLP method. First, the method will be used to evaluate the intervals of variation of the drain current introduced by RTN in the presence of noise, which can be helpful from a circuit design point of view [54]. As a second example, the method will be used to separate the RTN and BTI components in relaxation traces in CMOS devices after their stress, to facilitate the extraction of the required information for BTI compact modeling [55].

4.1 Evaluating the Intervals of Variations of the Drain Current in MOSFETs

As demonstrated previously, from a physics point of view, the w-TLP method is useful to obtain information on the properties of the defects involved in the RTN. In this section, we will show that it is also able to provide useful information from a circuit point of view such as the margins of variation of the drain current in a MOSFET, introduced by RTN. The knowledge of these variations during the operation of the device can help to establish precise safety margins for the circuit operation. However, as demonstrated before, the identification of the RTN current levels, which will affect the current limits, can be difficult because of the presence of a background noise, due to other active physical processes in the device or to experimental factors. This background noise, which is unknown in experimental RTN traces, can mask the current levels, especially if the background noise is large and the levels are very close [38]. Hereafter, we will show that the w-TLP can be used to obtain the rms value, σ , of the additional noise, improving the

Fig. 14 Monte

Carlo-generated RTN, which results from the charge fluctuations in two defects, so that four different current levels (black line) are observed. Gaussian noise with $\sigma = 0.4$ nA has been added to the RTN signal to reproduce the background noise (circles)

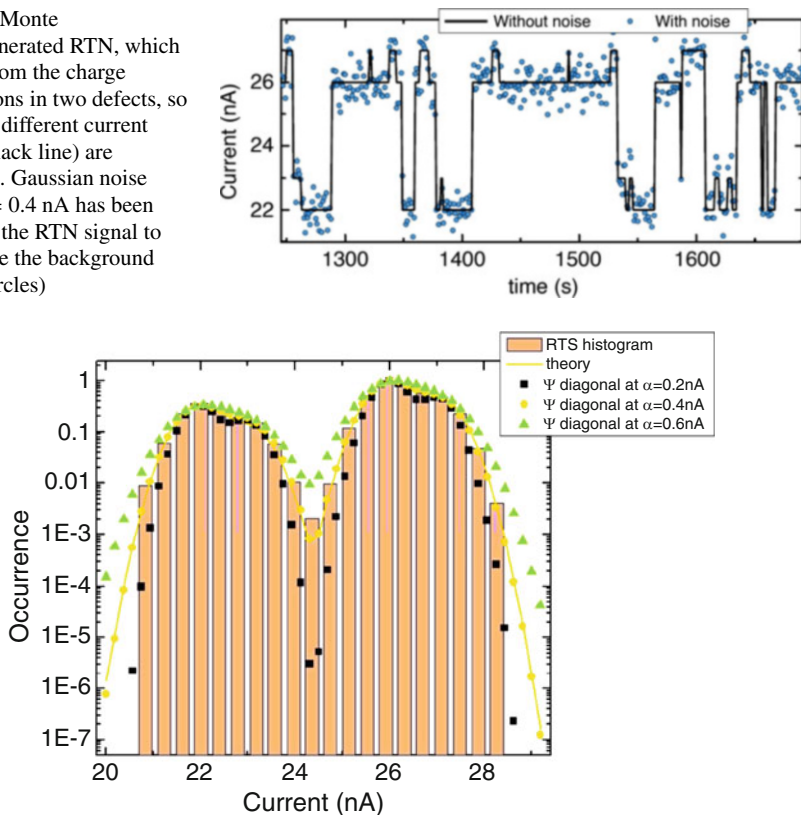


Fig. 15 Histogram of the currents in the RTN signal in Fig. 14 (bars). The yellow line corresponds to the theoretical $F(I)$ function (Eq. 3). The symbols show the w-TLP Ψ diagonal function with $\alpha = 0.2$ nA (black), 0.4 nA (yellow), and 0.6 nA (green)

evaluation of the amplitude of the RTN drain current fluctuations. As in Sect. 2, first, a numerically generated RTN signal will be considered as an example and, afterwards, RTN coming from experimental measurements in pMOS transistors will be studied.

Figure 14 illustrates a typical RTN signal, in this case numerically built. The black solid line in Fig. 14 shows a Monte Carlo-generated RTN signal caused by two defects, which leads to four current levels, centered at 22, 23, 26, and 27 nA. To get a more realistic RTN, a background Gaussian noise with standard deviation $\sigma = 0.4$ nA has been added (circles in Fig. 14).

The histogram of the RTN currents in Fig. 14 is plotted in Fig. 15 (bars). Since we are dealing with a numerically generated RTN, it is straightforward to deduce the analytical equation that describes this histogram. For each of the current levels of the generated RTN, a Gaussian distribution is built, with average $I_{L,i}$ (the value of each current level). The standard deviation is the same for all the distributions

and equal to the standard deviation of the background noise, i.e., $\sigma = 0.4$ nA. The height of each Gaussian distribution is proportional to the probability of occurrence of the associated current level. These probabilities can be easily calculated from the (known, in this case) input parameters used to generate the RTN. Therefore, the analytical probability density function obeys Eq. (3):

$$F(I) = \frac{K}{2\pi\sigma^2} \sum_{i=1}^L A_i \exp\left(\frac{-(I - I_{L,i})^2}{2\sigma^2}\right) \quad (3)$$

being L the number of levels in the RTN, i an index that denotes the level, I the current at each point of the RTN, σ the standard deviation of the background noise, A_i the peak height, and $I_{L,i}$ the current of each RTN level. “ K ” is a normalization constant, whose value has been chosen so that the maximum of $F(I)$ is 1. The red line in Fig. 15 corresponds to the analytical probability distribution obtained from Eq. (3). But in a realistic case, the parameters of the Gaussian distributions are unknown, so $F(I)$ cannot be built. The w-TLP method, however, can be used to fit the current histogram. To show this point, symbols in Fig. 15 show the Ψ functions, calculated according to Eq. (2) for the case $x = y$, i.e., the diagonal of the function, for different values of α . From now on, we will refer to this function as “the w-TLP Ψ diagonal function.” Note that the best match between the w-TLP Ψ diagonal function and the RTN histogram is obtained for $\alpha = 0.4$ nA, that is, when α is equal to the standard deviation of the background noise, σ .

Since σ is unknown in experimental RTN traces, these results indicate that the noise σ can be calculated using the w-TLP method, as the α value of the w-TLP Ψ diagonal function that better fits the RTN histogram. Additionally, the w-TLP Ψ diagonal function with $\alpha = \sigma$ (yellow circles) also matches the analytical probability density function (yellow line), even outside the histogram limits. This leads to a second conclusion: the w-TLP Ψ diagonal function that best fits the RTN histogram can be used to evaluate the intervals of variation of the drain current in the device, even outside the experimental window, that is, for longer operation times.

In the following, this methodology will be applied to determine the intervals of variation of the drain current from RTN traces experimentally obtained, where all the parameters that define the RTN parameters are unknown. As examples, Fig. 16 shows the RTN traces obtained in high-k/metal gate pMOS transistor biased at a gate voltage $V_G = -0.6$ V and drain voltages ranging from -50 mV (bottom) to -300 mV (top). The discrete sudden changes in the current reveal the presence of defects in the device, leading to RTN in the drain current.

As explained before, for a correct evaluation of the limits of the drain current fluctuations, the α value, which is indicative of the standard deviation of the background noise (σ) and unknown in experimental measurements, must be carefully chosen. To evaluate the suitable α , we use the property demonstrated in Fig. 15, that is, the best fit of the RTN histogram with the w-TLP Ψ diagonal function is obtained when $\alpha = \sigma$. So, the w-TLP Ψ diagonal function has been calculated for a wide

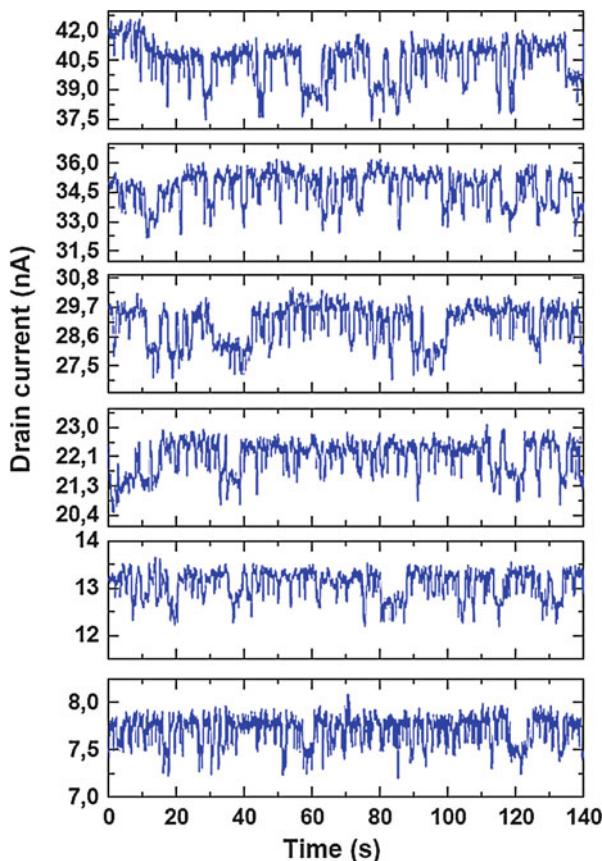


Fig. 16 RTN traces measured in the drain current of a high-k/metal gate pMOS transistor. Different drain voltages were applied, ranging from -50 mV (bottom) to -300 mV (top) with -50 mV intervals, with a gate voltage of -0.6 V

range of α values, and the difference between this function and the corresponding histograms (the error) has been calculated. As an example, Fig. 17 (top) shows the envelope of the histogram of the RTN obtained at $V_D = -250$ mV (dots). Figure 17 (bottom) shows the fitting error, as a function of the α values. The minimum of the error function in Fig. 17 (bottom) corresponds to $\alpha = \sigma$ (approximately 150 pA in this case). The corresponding w-TLP Ψ diagonal function has also been plotted in Fig. 17 (top), showing a good fit to the experimental data.

Figure 18 shows the envelopes of the current histograms of all the RTN traces in Fig. 16 and the corresponding best-fit curves. Extrapolating the w-TLP Ψ diagonal function to low occurrence probabilities provides the information about the expected lower and upper limits of the drain current variations caused by RTN for large operation times of the transistors [54]. Note that the function is very steep, leading to well-defined limits for the drain current.

Fig. 17 (top) Envelope of the current histogram of the RTN trace measured at $V_D = -250$ mV in Fig. 16 (dots). Best-fit w-TLP Ψ diagonal function (line). (Bottom) Error function of the fitting

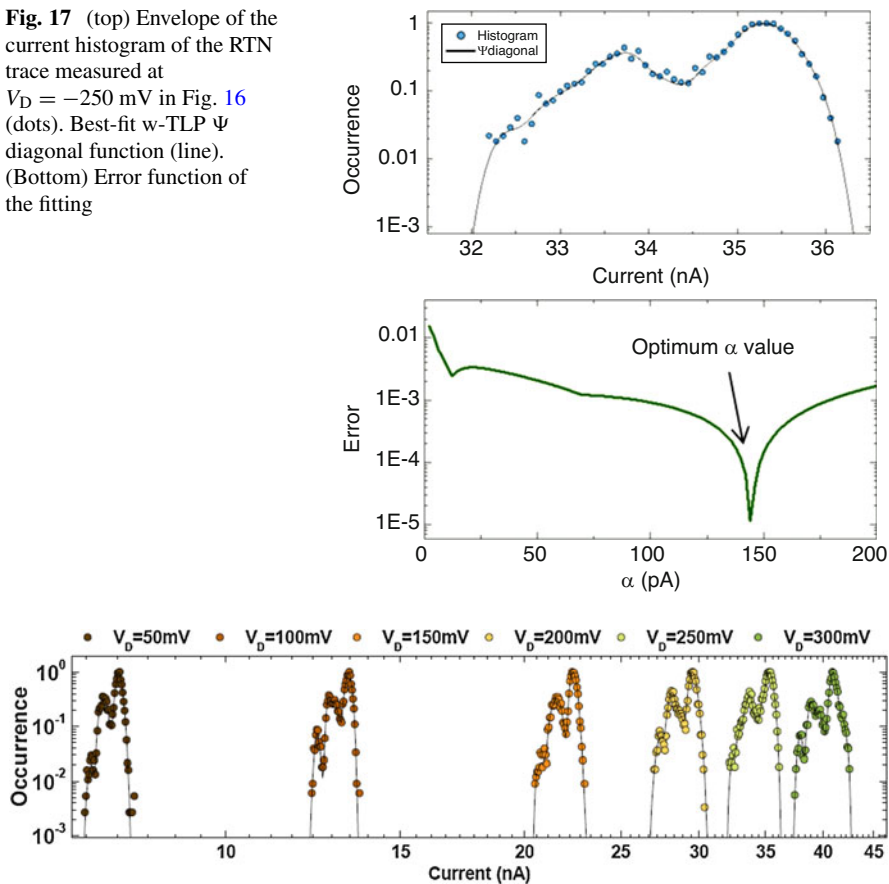


Fig. 18 Current histograms of the RTN traces in Fig. 16 (symbols) and their best-fit w-TLP Ψ diagonal function (lines). The extrapolation to low-probability occurrence allows determining the upper and lower bounds of the drain current

4.2 RTN and BTI Phenomena Identification in CMOS Devices

As described in Sect. 1, nowadays it is accepted that RTN and BTI are two manifestations of the same phenomenon (i.e., the trapping/detrapping of charges in/from defects in the device), being the main difference the time constants of the defects involved, faster in the case of RTN. BTI is expected at larger voltages and consists of a permanent and a recovery component. For the characterization of the aging, the recovery component is usually analyzed, from which the defect parameters can be evaluated. Actually, aging can be modeled by analyzing the emission times (τ_e) and related $\Delta V_{th}(\eta)$ of defects during the recovery phases, after the application of overvoltage stress [35]. Figure 19a shows a typical experimental recovery trace after a BTI test, after the conversion of the measured current into the

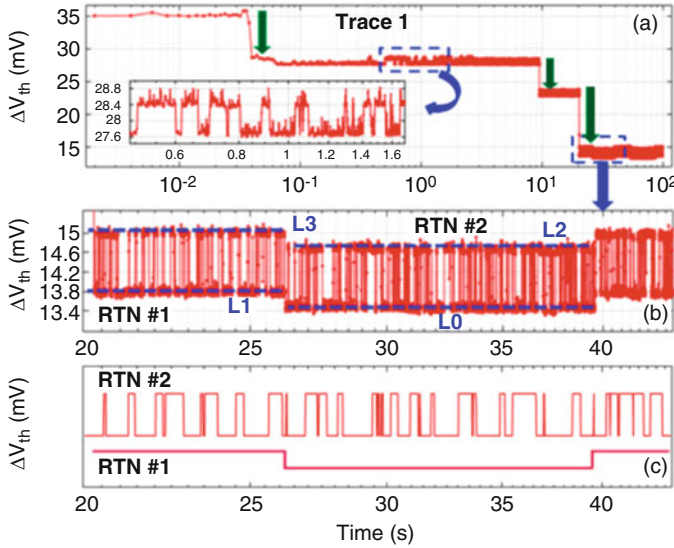


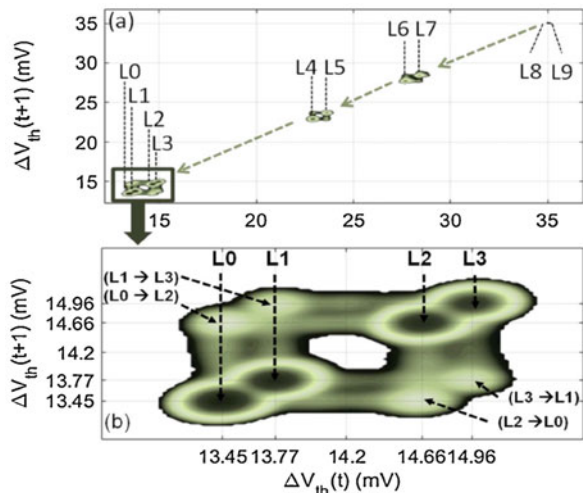
Fig. 19 (a) Resulting ΔV_{th} trace from the measured I_{DS} recovery data showing three clear slow defect discharges. The inset in figure (a) shows RTN and background noise on the top of a ΔV_{th} level. (b) Zoom of 23 s where four different ΔV_{th} levels can be distinguished corresponding to two fast defects charge/discharges. (c) Illustration of the RTN signals in (b) uncoupled

equivalent threshold voltage change [56]. Charge detrapping from oxide defects can be clearly seen during the recovery period as abrupt current jumps. The zoom in Fig. 19b shows the simultaneous presence of RTN in the traces. Depending on the amplitude of the RTN, the net current changes related to the BTI aging component could be masked, so, for the correct BTI modeling, it is critical to distinguish between current changes that are linked to BTI or to RTN. With this purpose, the w-TLP method will be used. Note that BTI is also a stochastic process, so that many traces as that shown in Fig. 19a will have to be analyzed to get statistically meaningful information.

As an example, the trace in Fig. 19a is considered. In this figure, three slow emissions (denoted with green arrows) without any capture counterpart are observed, linked to BTI and whose associated τ_e and η must be evaluated. However, a detailed inspection reveals that RTN is superposed to some of the levels. The inset in Fig. 19a exposes fast transitions between two separated ΔV_{th} levels on top of a slow emission. Figure 19b reveals fast capture/emission transitions mixed with background noise, switching between four different ΔV_{th} levels (i.e., L0–L3). A visual inspection of Fig. 19b reveals the join contribution of two individual RTN signals, which have been separated in Fig. 19c in order to visually distinguish them.

The w-TLP method allows the identification of the fast and slow defects, by identifying the number, i.e., L0 to $\#V_{th-LEVELS}-1$ and value of the ΔV_{th} shifts present in the trace. Figure 20a shows the w-TLP resulting from the trace in Fig. 19a. By

Fig. 20 (a) w-TLP resulting from the analysis of Trace #1 in Fig. 19a. (b) Zoom of the region marked with a red square in the w-TLP in (a) showing, for the two RTN signals in Fig. 19b, the ΔV_{th} levels in the diagonal and the corresponding transitions between ΔV_{th} levels outside the diagonal



analyzing the counter diagonal of the w-TLP, four groups of populated data regions, separated in the figure by green dashed arrows, can be distinguished. Each of the data groups corresponds to a different ΔV_{th} level present in the recovery trace. Transitions from one data group to the next one are considered as slow emissions at a specific τ_e and the difference between the two levels corresponds to the η value of the associated defect.

The fast capture/emissions on top of each ΔV_{th} level are clearly distinguished as red populated regions in the counter diagonal (i.e., ΔV_{th} levels L0–L9) with other less-populated regions outside the counter diagonal (i.e., ΔV_{th} -level transitions). Figure 20b shows a zoom in of the w-TLP, between $\Delta V_{th}(t) = 13$ mV and $\Delta V_{th}(t + 1) = 15$ mV. The zoom clearly shows four different ΔV_{th} levels resulting from the combination of the charge/discharges of two fast defects, shown in the zoom of “Trace #1” in Fig. 19c as individual RTN signals:

- RTN #1: As shown in Fig. 19b, the two associated main ΔV_{th} levels are located at L0 and L1. Transitions between these two ΔV_{th} levels occur from level L0 to L1 and vice versa.
- RTN #2: In this case, the two main ΔV_{th} levels are located always above the RTN #1 level. For instance, at recovery time of 25 s, RTN #1 stays at level L1, thus the RTN #2 switches between levels L1 and L3. On the contrary, at the recovery time of 30 s, the RTN #1 has dropt to level L0 and the RTN #2 switches between L0 and L2.

Once the ΔV_{th} levels are identified by the w-TLP, the method assigns, to each sample in the ΔV_{th} trace, the closest ΔV_{th} level obtained with the w-TLP. This step quantifies the ΔV_{th} recovery trace, removing the background noise and leaving only ΔV_{th} levels associated to slow and fast defects capture/emissions. From the clean traces, η and τ_e related to the BTI aging component in the trace can be obtained,

for the modeling of the RTN phenomenon [52]. It must be emphasized that the complete process can be automatized, to evaluate the large number of traces needed for accurate statistical analysis.

5 Conclusions

In this chapter, some advanced RTN characterization techniques have been presented. On the one hand, some of the solutions proposed to overcome the main experimental limitations linked to the characterization of RTN have been revised. First, since a statistical characterization of the phenomenon is needed, an array-based chip has been proven as an area-efficient solution, which can partially contribute to the reduction of the testing time. Second, a new experimental setup has been presented, which provides a higher time resolution than standard equipment, so that faster trapping/detrapping events can be captured. To analyze the obtained data, the weighted time lag plot (w-TLP) method has been presented, being an efficient and robust method to evaluate the defect-related parameters, even when the background noise is large. The potentiality of the method has been demonstrated through several examples: RTN analysis in RRAM and CMOS devices, the evaluation of the statistical distributions of the drain current in MOSFETs in the presence of RTN and the identification of slow emissions and RTN transients during BTI testing.

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References

1. M.J. Kirton, M.J. Uren, Noise in solid-state microstructures: a new perspective on individual defects, interface states and low-frequency ($1/f$) noise. *Adv. Phys.* **38**(4), 367–468 (1989)
2. K.R. Farmer, Discrete conductance fluctuations and related phenomena in metal-oxide-silicon device structures, in *Proc. Insulating Films on Semiconductors*, 1991, pp. 1–18
3. E. Simoen, C. Claeys, Random telegraph signals in silicon-on-insulator metal-oxide-semiconductor transistors. *J. Appl. Phys.* **75**(7), 3647–3653 (1994)
4. K.S. Ralls, W.J. Skocpol, L.D. Jackel, R.E. Howard, L.A. Fetter, R.W. Epworth, D.M. Tennant, Discrete resistance switching in submicrometer silicon inversion layers: individual interface traps and ($1/f$?) noise. *Phys. Rev. Lett.* **52**(3), 228–231 (1984)
5. R. Huang, X.B. Jiang, S.F. Guo, P.P. Ren, P. Hao, Z.Q. Yu, Z. Zhang, Y.Y. Wang, R.S. Wang, Variability-and reliability-aware design for 16/14nm and beyond technology, in *2017 IEEE International Electron Devices Meeting (IEDM)*, pp. 12.4.1–12.4.4
6. S.S. Chung, Recent advances of RTN technique towards the understanding of the gate dielectric reliability in trigate FinFETs, in *2016 IEEE 23rd International Symposium on the Physical and Failure Analysis of Integrated Circuits (IPFA)*, 2016, pp. 33–37
7. C.G. Theodorou, E.G. Ioannidis, S. Haendler, N. Planes, E. Josse, C.A. Dimitriadis, G. Ghibaudo, New LFN and RTN analysis methodology in 28 and 14nm FD-SOI MOSFETs, in *2015 IEEE International Reliability Physics Symposium*, pp. XT.1.1–XT.1.6

8. C. Marquez, N. Rodriguez, F. Gamiz, A. Ohata, Electrical characterization of random telegraph noise in Back-biased ultrathin silicon-on-insulator MOSFETs, in *Eurosoi-Ulis*, 2016
9. C. Claeys, M.G.C. de Andrade, Z. Chai, W. Fang, B. Govoreanu, B. Kaczer, W. Zhang, E. Simoen, Random telegraph signal noise in advanced high performance and memory devices, in *31st Symposium on Microelectronics Technology and Devices (SBMicro)*, 2016
10. Y. Zhang, X. Jiang, J. Wang, S. Guo, Y. Fang, R. Wang, M. Luo, R. Huzng, Impacts of random telegraph noise (RTN) on the energy-delay tradeoffs of logic circuits, in *2016 China Semiconductor Technology International Conference (CSTIC)*, pp.1–4
11. K. Ito, T. Matsumoto, S. Nishizawa, H. Sunagawa, K. Kobayashi H. Onodera, The impact of RTN on performance fluctuation in CMOS logic circuits, in *2011 International Reliability Physics Symposium*, pp. CR.5.1–CR.5.4
12. D.S. Yaney, C.Y. Lu, R.A. Kohler, M.J. Kelly, J.T. Nelson, A meta-stable leakage phenomenon in DRAM charge storage—variable hold time, in *Tech. Dig. IEDM*, 1987, pp. 336–339
13. P.J. Restle, J.W. Park, B.F. Floyd, DRAM variable retention time, in *Tech. Dig. IEDM*, 1992, pp. 807–810
14. H. Qiu et al., Impact of random telegraph noise on write stability in silicon-on-thin-BOX (SOTB) SRAM cells at low supply voltage in sub-0.4V regime, in *VLSI Technology*, 2015
15. A. Goda, C. Miccoli, C. Monzio Compagnoni, Time dependent threshold-voltage fluctuations in NAND flash memories: from basic physics to impact on array operation, in *Tech. Dig. IEDM*, 2015
16. B. Zimmer, O. Thomas, S. Oon Toh, T. Vincent, K. Asanovi'c, B. Nikoli'c, Joint impact of random variations and RTN on dynamic writeability in 28nm bulk and FDSOI SRAM, in *44th European Solid State Device Research Conference (ESSDERC)*, 2014
17. K.C. Akyel, L. Ciampolini, O. Thomas, D. Turgis, G. Ghibaudo, UTBB FD-SOI front- and back-gate coupling aware random telegraph signal impact analysis on a 6T SRAM, in *SOI-3D-Subthreshold Microelectronics Technology Unified Conference (S3S)*, 2014
18. J.P. Noel et al., Multi-VT UTBB FDSOI device architectures for low-power CMOS circuit. *IEEE Trans. Electron Devices* **58**(8), 2473–2482 (2011)
19. H. Akinaga, H. Shima, Resistive random access memory (RRAM) based on metal oxides. *Proc. IEEE* **98**(12), 2237–2251 (2010)
20. G. Burr, B. Kurdi, J. Scott, C. Lam, K. Gopalakrishnan, R. Shenoy, Overview of candidate device technologies for storage-class memory. *IBM J. Res. Dev.* **52**(4/5), 449–464 (2008)
21. J.-K. Lee, J.-W. Lee, J. Park, S.-W. Chung, J.S. Roh, S.-J. Hong, I. Cho, H.-I. Kwon, J.-H. Lee, Extraction of trap location and energy from random telegraph noise in amorphous TiOx resistance random access memories. *Appl. Phys. Lett.* **98**, 143502-1–143502-3 (2011)
22. F. Puglisi, Random telegraph noise analysis as a tool to link physical device features to electrical reliability in nanoscale devices, in *IIRW*, 2016
23. A.S.M. Shamsur Rouf, Z. Çelik-Butler, *Oxide Trap-Induced RTS in MOSFETs*, Chapter 17 of this book, 2019
24. C. Claeys et al., Random telegraph signal noise in advanced high performance and memory devices, in *Symposium on Microelectronics Technology and Device*, 2016, pp. 1–6
25. A. Yonezawa, A. Teramoto, T. Obara, R. Kuroda, S. Sugawa, T. Ohmi, The study of time constant analysis in random telegraph noise at the subthreshold voltage region, in *2011 IEEE International Conference on IC Design & Technology*, pp. 1–4
26. V. Velayudhan, J. Martin-Martinez, R. Rodriguez, M. Porti, M. Nafria, X. Aymerich, C. Medina, F. Gamiz, TCAD simulation of interface traps related variability in bulk decananometer MOSFETs, in *2014 5th European Workshop on CMOS Variability (VARI)*, pp. 1–6
27. B. Kaczer, S. M. Amoroso, R. Hussin, A. Asenov, J. Franco, P. Weckx, Ph.J. Roussel, G. Rzepa, T. Grasser, N. Horiguchi, On the distribution of the FET threshold voltage shifts due to individual charged gate oxide defects, in *2016 IEEE International Integrated Reliability Workshop (IIRW)*, pp. 18–20
28. F. Adamu-Lema, C. Compagnoni, O. Badami, V. Georgiev, A. Asenov, *RTN and Its Intrinsic Interaction with Statistical Variability Sources in Advanced Nanoscale Devices: A Simulation Study*, Chapter 13 of this book, 2019

29. E. Simoen, C. Claeys, Random telegraph signal: a local probe for single point defect studies in solid-state devices. *Mater. Sci. Eng. B* **91–92**, 136–143 (2002)
30. J. Franco et al., Impact of single charged gate oxide defects on the performance and scaling of nanoscaled FETs, in *Proc. IEEE Int. Rel. Phys. Symp. (IRPS)*, Apr. 2012, pp. 5A.4.1–5A.4.6
31. M. Toledano-Luque, B. Kaczer, J. Franco, Ph.J. Roussel, M. Bina, T. Grasser, M. Cho, P. Weckx, G. Groeseneken, Degradation of time dependent variability due to interface state generation, in *2013 Symposium on VLSI Technology*, pp. T190–T191
32. N. Ayala, J. Martin-Martinez, R. Rodriguez, M. Nafria, X. Aymerich, Unified characterization of RTN and BTI for circuit performance and variability simulation, in *2012 Proceedings of the European Solid-State Device Research Conference (ESSDERC)*, pp. 266–269
33. M. Toledano-Luque, B. Kaczer, E. Simoen, R. Degraeve, J. Franco, Ph.J. Roussel, T. Grasser, G. Groeseneken, Correlation of single trapping and detrapping effects in drain and gate currents of nanoscaled nFETs and pFETs, in *2012 IEEE International Reliability Physics Symposium (IRPS)*, pp. XT.5.1–XT.5.6
34. T. Grasser, K. Rott, H. Reisinger, M. Walzl, J. Franco, B. Kaczer, A unified perspective of RTN and BTI, in *2014 IEEE International Reliability Physics Symposium*, pp. 4A.5.1–4A.5.7
35. J. Martin-Martinez, B. Kaczer, M. Toledano-Luque, R. Rodriguez, M. Nafria, X. Aymerich, G. Groeseneken, Probabilistic defect occupancy model for NBTI, in *2011 International Reliability Physics Symposium*, pp. XT.4.1–XT.4.6
36. T. Nagumo, K. Takeuchi, S. Yowogawa et al., New analysis methods for comprehensive understanding of random telegraph noise, in *Tech. Dig. IEDM*, Dec. 2009, pp. 32.1.1–32.1.4
37. C. Monzio Compagnoni, R. Gusmeroli, A.S. Spinelli, et al., Statistical model for random telegraph noise in flash memories. *IEEE Trans. Electron Devices* **55**(1), 388–395 (2008)
38. Y. Yuzhelevski, M. Yuzhelevski, G. Jung, Random telegraph noise analysis in time domain. *Rev. Sci. Instrum.* **71**(4), 1681–1688 (2000)
39. J. Martin-Martinez, J. Diaz, R. Rodriguez, M. Nafria, X. Aymerich, New weighted time lag method for the analysis of random telegraph signals. *IEEE Electron Device Lett.* **35**(4), 479–481 (2014)
40. K. Abe, A. Teramoto, S. Shigetoshim et al., Understanding of traps causing random telegraph noise based on experimentally extracted time constants and amplitude, in *Proc. IRPS*, 2011, pp. 4A.4.1–4A.4.6
41. C. Marquez, O. Huerta, A.I. Tec-Chim, F. Guarin, E.A. Gutierrez-D, F. Gamiz, *Systematic Characterization of Random Telegraph Noise and Its Dependence with Magnetic Fields in MOSFET Devices*, Chapter 4 of this book, 2019
42. <https://www.tek.com/sites/default/files/media/media/resources/4225-PMUDataSht.pdf>
43. <https://www.keysight.com/en/pd-1443698-pn-B1500A-A30re/waveform-generator-fast-measurement-unit-wgfm-module-for-the-b1500a?cc=ES&lc=eng>
44. N. Raghavan, R. Degraeve, A. Fantini, L. Goux, S. Strangio, B. Govoreanu, D.J. Wouters, G. Groeseneken, M. Jurczak, Microscopic origin of random telegraph noise fluctuations in aggressively scaled RRAM and its impact on read disturb variability, in *IEEE International Reliability Physics Symposium (IRPS)*, 2013, pp. 5E.3.1–5E.3.7
45. F.M. Puglisi, P. Pavan, L. Larcher, A. Padovani, Analysis of RTN and cycling variability in HfO₂ RRAM devices in LRS, in *European Solid State Device Research Conference (ESSDERC)*, 2014, pp. 246–249
46. D. Ielmini, F. Nardi, C. Cagli, Resistance-dependent amplitude of random telegraph-signal noise in resistive switching memories. *Appl. Phys. Lett.* **96**(5), 053503_1–053503_3 (2010)
47. H.-S.P. Wong, H.-Y. Lee, S. Yu, Y.-S. Chen, Y. Wu, P.-S. Chen, B. Lee, F.T. Chen, M.-J. Tsai, Metal-oxide RRAM. *Proc. IEEE* **100**(6), 1951–1970 (2012)
48. C.S. Chen, L. Li, Q. Lim, H.H. Teh, N.F.B. Omar, C.L. Ler, J.T. Watt, A compact array for characterizing 32k transistors in wafer scribe lanes, in *2014 IEEE Conference on Microelectronic Test Structures*, pp. 227–232
49. P. Weckx, B. Kaczer, C. Chen, P. Raghavan, D. Linten, A. Mocuta, Relaxation of time-dependent NBTI variability and separation from RTN, in *Proc. IRPS*, 2017

50. P. Weckx, B. Kaczer, C. Chen, J. Franco, E. Bury, K. Chanda, J. Watt, Ph.J. Roussel, F. Catthoor, G. Groeseneken, Characterization of time-dependent variability using 32k transistor arrays in an advanced HK/MG technology, in *Proc. IRPS*, 2015
51. A. Whitcombe, S. Taylor, M. Denham, V. Milovanovic, B. Nikoli, On-chip I-V variability and random telegraph noise characterization in 28 nm CMOS, in *46th European Solid-State Device Research Conference (ESSDERC)*, 2016, pp. 248–251
52. J. Diaz-Fortuny, J. Martin-Martinez, R. Rodriguez, M. Nafria, R. Castro-Lopez, E. Roca, F.V. Fernandez, E. Barajas, X. Aragones, D. Mateo, A transistor array chip for the statistical characterization of process variability, RTN and BTI/CHC aging, in *14th International Conference on Synthesis, Modeling, Analysis and Simulation Methods and Applications to Circuit Design (SMACD)*, 2017
53. J. Diaz-Fortuny, J. Martin-Martinez, R. Rodriguez, M. Nafria, R. Castro-Lopez, E. Roca, F.V. Fernandez, TARS: a toolbox for statistical reliability modeling of CMOS devices, in *14th International Conference on Synthesis, Modeling, Analysis and Simulation Methods and Applications to Circuit Design (SMACD)*, 2017
54. J. Martin-Martinez, J. Diaz, R. Rodriguez, M. Nafria, X. Aymerich, E. Roca, F.V. Fernandez, A. Rubio, Characterization of random telegraph noise and its impact on reliability of SRAM sense amplifiers, in *2014 5th European Workshop on CMOS Variability (VARI)*
55. J. Diaz-Fortuny, J. Martin-Martinez, R. Rodriguez, M. Nafria, R. Castro-Lopez, E. Roca, F.V. Fernandez, A noise and RTN-removal smart method for parameters extraction of CMOS aging compact models, in *2018 Joint International EUROSOI Workshop and International Conference on Ultimate Integration on Silicon (EUROSOI-ULIS)*
56. B. Kaczer et al., Ubiquitous relaxation in BTI stressing-new evaluation and insights, in *IEEE Int. Reliab. Phys. Symp. Proc.*, 2008, pp. 20–27